Q. No.	Subject	Chapter	Topic
1	Physics	Electrostatics	Electrostatics
2	Physics	Electrostatics	Electrostatics
3	Physics	Electrostatics	Electrostatics
4	Physics	Electrostatics	Electrostatics
5	Physics	Electrostatics	Electrostatics
6	Physics	Electrostatics	Electrostatics
7	Physics	Electrostatics	Electrostatics
8	Physics	Electrostatics	Electrostatics
9	Physics	Electrostatics	Electrostatics
10	Physics	Electrostatics	Electrostatics
11	Physics	Electrostatics	Electrostatics
12	Physics	Electrostatics	Electrostatics
13	Physics	Electrostatics	Electrostatics
14	Physics	Electrostatics	Electrostatics
15	Physics	Electrostatics	Electrostatics
16	Physics	Electrostatics	Electrostatics
17	Physics	Electrostatics	Electrostatics
18	Physics	Electrostatics	Electrostatics
19	Physics	Electrostatics	Electrostatics
20	Physics	Electrostatics	Electrostatics
21	Physics	Electrostatics	Electrostatics
22	Physics	Electrostatics	Electrostatics
23	Physics	Electrostatics	Electrostatics
24	Physics	Electrostatics	Electrostatics
25	Physics	Electrostatics	Electrostatics
26	Chemistry	Solutions	Solutions
27	Chemistry	Solutions	Solutions
28	Chemistry	Solutions	Solutions

29	Chemistry	Solutions	Solutions
30	Chemistry	Solutions	Solutions
31	Chemistry	Solutions	Solutions
32	Chemistry	Solutions	Solutions
33	Chemistry	Solutions	Solutions
34	Chemistry	Solutions	Solutions
35	Chemistry	Solutions	Solutions
36	Chemistry	Solutions	Solutions
37	Chemistry	Solutions	Solutions
38	Chemistry	Solutions	Solutions
39	Chemistry	Solutions	Solutions
40	Chemistry	Solutions	Solutions
41	Chemistry	Solutions	Solutions
42	Chemistry	Solutions	Solutions
43	Chemistry	Solutions	Solutions
44	Chemistry	Solutions	Solutions
45	Chemistry	Solutions	Solutions
46	Chemistry	Solutions	Solutions
47	Chemistry	Solutions	Solutions
48	Chemistry	Solutions	Solutions
49	Chemistry	Solutions	Solutions
50	Chemistry	Solutions	Solutions
51	Maths	Matrices & Determinants	Matrices & Determinants
52	Maths	Matrices & Determinants	Matrices & Determinants
53	Maths	Matrices & Determinants	Matrices & Determinants
54	Maths	Matrices & Determinants	Matrices & Determinants
55	Maths	Matrices & Determinants	Matrices & Determinants
56	Maths	Matrices & Determinants	Matrices & Determinants
57	Maths	Matrices & Determinants	Matrices & Determinants
58	Maths	Matrices & Determinants	Matrices & Determinants
59	Maths	Matrices & Determinants	Matrices & Determinants
60	Maths	Matrices & Determinants	Matrices & Determinants
61	Maths	Matrices & Determinants	Matrices & Determinants

62	Maths	Matrices & Determinants	Matrices & Determinants
63	Maths	Matrices & Determinants	Matrices & Determinants
64	Maths	Matrices & Determinants	Matrices & Determinants
65	Maths	Matrices & Determinants	Matrices & Determinants
66	Maths	Matrices & Determinants	Matrices & Determinants
67	Maths	Matrices & Determinants	Matrices & Determinants
68	Maths	Matrices & Determinants	Matrices & Determinants
69	Maths	Matrices & Determinants	Matrices & Determinants
70	Maths	Matrices & Determinants	Matrices & Determinants
71	Maths	Matrices & Determinants	Matrices & Determinants
72	Maths	Matrices & Determinants	Matrices & Determinants
73	Maths	Matrices & Determinants	Matrices & Determinants
74	Maths	Matrices & Determinants	Matrices & Determinants
75	Maths	Matrices & Determinants	Matrices & Determinants

Subtopic Name	Question Type
COULOMB'S LAW AND VECTOR NOTATION	MCQ
ELECTRIC FIELD INTENSITY	MCQ
Closest matching subtopic: SUPERPOSITION OF CHARGES - SHM/time-period part not separately listed	MCQ
SUPERPOSITION OF CHARGES	MCQ
COULOMB'S LAW AND VECTOR NOTATION	MCQ
Closest matching subtopic: COULOMB'S LAW AND VECTOR NOTATION - mechanical equilibrium with tension not separately listed	MCQ
Closest matching subtopic: APPLICATIONS OF GAUSS'S LAW (CYLINDRICAL AND SPHERICAL CHARGE DISTRIBUTION) - exact charge-density integration subtopic not found	MCQ
ELECTROSTATIC POTENTIAL ENERGY	MCQ
SUPERPOSITION OF CHARGES	MCQ
SUPERPOSITION OF CHARGES	MCQ
APPLICATIONS OF GAUSS'S LAW (LINE & SURFACE CHARGE DISTRIBUTION)	MCQ
Closest matching subtopic: ELECTRIC CHARGE AND ITS PROPERTIES - Millikan oil-drop experiment not separately listed	MCQ
APPLICATIONS OF GAUSS'S LAW (LINE & SURFACE CHARGE DISTRIBUTION)	MCQ
ELECTRIC FIELD INTENSITY	MCQ
ELECTROSTATIC POTENTIAL ENERGY	MCQ
Closest matching subtopic: ELECTRIC FIELD INTENSITY - charged-particle trajectory not separately listed	MCQ
SUPERPOSITION OF CHARGES	MCQ
SUPERPOSITION OF CHARGES	MCQ
SUPERPOSITION OF CHARGES	MCQ
SUPERPOSITION OF CHARGES	MCQ
SUPERPOSITION OF CHARGES	Numerical/Integer
Closest matching subtopic: COULOMB'S LAW AND VECTOR NOTATION - incline equilibrium not separately listed	Numerical/Integer
SUPERPOSITION OF CHARGES	Numerical/Integer
CONDUCTION, INDUCTION & FRICTION	Numerical/Integer
COULOMB'S LAW AND VECTOR NOTATION	Numerical/Integer
EXPRESSING CONCENTRATION OF SOLUTIONS	MCQ
EXPRESSING CONCENTRATION OF SOLUTIONS	MCQ
SOLUBILITY	MCQ

SOLUBILITY	MCQ
RAOULT'S LAW	MCQ
EXPRESSING CONCENTRATION OF SOLUTIONS	Assertion-Reason
RAOULT'S LAW	MCQ
IDEAL AND NON-IDEAL SOLUTIONS	MCQ
EXPRESSING CONCENTRATION OF SOLUTIONS	Statement I/II
RAOULT'S LAW	Assertion-Reason
IDEAL AND NON-IDEAL SOLUTIONS	Assertion-Reason
IDEAL AND NON-IDEAL SOLUTIONS	MCQ
EXPRESSING CONCENTRATION OF SOLUTIONS	Match the following
SOLUBILITY	Match the following
IDEAL AND NON-IDEAL SOLUTIONS	Match the following
Closest matching subtopic: EXPRESSING CONCENTRATION OF SOLUTIONS - general physical-state classification of solutions not found	MCQ
RAOULT'S LAW	MCQ
VAPOUR PRESSURE OF SOLUTIONS	MCQ
SOLUBILITY	MCQ
EXPRESSING CONCENTRATION OF SOLUTIONS	MCQ
EXPRESSING CONCENTRATION OF SOLUTIONS	Numerical/Integer
RAOULT'S LAW	Numerical/Integer
EXPRESSING CONCENTRATION OF SOLUTIONS	Numerical/Integer
EXPRESSING CONCENTRATION OF SOLUTIONS	Numerical/Integer
SOLUBILITY	Numerical/Integer
DETERMINANTS	MCQ
Closest matching subtopic: ADDITION OF MATRICES - basic matrix formation/trace not separately listed	MCQ
TRANSPOSE OF MATRIX,	MCQ
MULTIPLICATION OF MATRICES	MCQ
MULTIPLICATION OF MATRICES	MCQ
Closest matching subtopic: ADDITION OF MATRICES - basic matrix formation/trace not separately listed	MCQ
SYMMETRIC AND SKEW SYMMETRIC MATRICES	MCQ
MULTIPLICATION OF MATRICES	MCQ
TRANSPOSE OF MATRIX,	MCQ
MULTIPLICATION OF MATRICES	MCQ
MULTIPLICATION OF MATRICES	MCQ

MULTIPLICATION OF MATRICES	MCQ
DETERMINANTS	MCQ
DETERMINANTS	MCQ
DETERMINANTS	MCQ
SYMMETRIC AND SKEW SYMMETRIC MATRICES	MCQ
MULTIPLICATION OF MATRICES	MCQ
DETERMINANTS	MCQ
DETERMINANTS	MCQ
SYSTEM OF HOMOGENEOUS EQUATIONS	MCQ
SYSTEM OF NON HOMOGENEOUS EQUATIONS	Numerical/Integer
DETERMINANTS	Numerical/Integer
MULTIPLICATION OF MATRICES	Numerical/Integer
Closest matching subtopic: DETERMINANTS - Cayley-Hamilton/characteristic equation not separately listed	Numerical/Integer
DETERMINANTS	Numerical/Integer

Marks	Level	Reason for Level
4	L1	Direct Coulomb-law substitution with unit conversion.
4	L1	Single formula E = kq/r^2 with simple powers.
4	L3	Needs restoring-force derivation and SHM time-period linkage.
4	L2	Requires resolving force from two charges and maximizing with x.
4	L2	Uses vector resultant of two equal electrostatic forces in equilateral geometry.
4	L2	Combines Coulomb force, tension balance and small-angle approximation.
4	L2	Requires integrating spherical charge density and solving logarithmically.
4	L1	Direct work by uniform electric field from displacement component.
4	L1	Two collinear Coulomb-force contributions added directly.
4	L2	Needs vector components of field from three square-corner charges.
4	L2	Requires field of uniformly charged circular arc using symmetry/integration.
4	L2	Combines density, weight-electric balance and charge quantization.
4	L2	Uses electric field due to finite uniformly charged wire with geometry.
4	L1	Direct use of F = qE with microcoulomb conversion.
4	L2	Requires work-energy integration in a position-dependent electric field.
4	L2	Combines uniform-field acceleration with projectile-style path after exit.
4	L1	Fields from two opposite charges at midpoint add along one line.
4	L2	Requires component-wise superposition with signed charges on a circle.
4	L2	Vector electric field from two off-axis charges with components.
4	L2	Needs equilibrium condition for all three collinear charges.
4	L2	Net force requires unequal-distance Coulomb forces and subtraction.
4	L2	Combines Coulomb repulsion with component of weight on incline.
4	L2	Requires maximizing force on perpendicular bisector using calculus.
4	L1	Contact charge sharing followed by direct Coulomb force calculation.
4	L1	Two perpendicular Coulomb forces combined by Pythagorean resultant.
4	L1	Direct molarity from density and mass percentage.
4	L1	Simple ppm-to-moles conversion plus K2SO4 gives 2K+.
4	L1	Direct Henry-law mole fraction with moles of water.

4 L2	Henry law, mole fraction and mass-per-litre conversion are combined.
4 L2	Requires two Raoult-law equations before solving pure vapour pressures.
4 L1	Molarity and molar mass calculation directly verifies assertion and reason.
4 L2	Compares vapour-phase and liquid-phase mole-fraction ratios via Raoult law.
4 L0	Direct conceptual fact: negative deviation lowers vapour pressure and raises boiling point.
4 L0	Direct definition of mole fraction and its sum.
4 L1	Uses mathematical form of Raoult's law and Henry's law.
4 L0	Direct ideal-solution property: enthalpy of mixing is zero.
4 L0	Direct concept of positive-deviation minimum-boiling azeotrope.
4 L0	Direct matching of concentration-term definitions.
4 L0	Direct matching of gas-solubility/Henry-law applications.
4 L0	Direct matching of ideal/non-ideal-solution properties and Henry law.
4 L0	Direct recognition of solution type example.
4 L1	Direct Raoult-law relation using solvent mole fraction.
4 L1	Compares given total vapour-pressure expression with Raoult-law form.
4 L2	Converts molality to mole fraction, then applies Henry's law.
4 L2	Requires relation between molarity, molality, density and molar mass.
4 L1	Uses molarity, solvent density and molar mass for molality.
4 L2	Uses vapour-liquid equilibrium relation yA P = xA PA degrees.
4 L2	Converts molality to molarity using density and solution mass.
4 L1	Direct molarity from given mass, molar mass and volume.
4 L1	Counts correct Henry-law statements by definition and trend.
4 L2	Determinant manipulation produces a quadratic relation for lambda.
4 L1	Trace reduces to summing diagonal terms with a standard square-sum formula.
4 L2	Requires computing a polynomial in A and then taking transpose.
4 L2	Uses matrix-vector multiplication equations to determine entries of A.
4 L3	Uses matrix powers/inverse-like cancellation before solving a quadratic-root expression.
4 L0	Direct substitution in a_ij formula.
4 L2	Uses symmetry, determinant condition and trace relation together.
4 L3	Needs recognition of transformed powers and high power of a Jordan-type matrix.
4 L2	Uses AAT = 9I to form equations in a and b.
4 L2	Uses A^2 = I constraints plus trace and determinant.
4 L1	Low-order powers of the given matrix directly test each option.

4	L2	Requires recognizing power pattern of triangular/nilpotent matrix.
4	L2	Row operations reduce determinant to polynomial coefficients.
4	L2	Column operations and determinant degree reasoning are needed.
4	L2	Uses nilpotency B^2 = 0 and determinant of simplified matrix expression.
4	L2	Needs parity of powers of skew-symmetric matrices under transpose.
4	L2	Uses power formula for upper triangular matrix with parameters a,b.
4	L2	Requires determinant zero and interval filtering.
4	L3	Multi-step determinant relation using given quadratic and bilinear constraints.
4	L2	Matrix power is simplified before solving a homogeneous system.
4	L2	Simplifies matrix powers then solves a non-homogeneous 2x2 system.
4	L3	Extremal determinant of 0-1 matrix requires non-routine reasoning.
4	L2	Needs matrix-power recurrence and use of one entry to find another.
4	L2	Uses characteristic equation/Cayley-Hamilton form for A.
4	L2	Uses logarithm identities and determinant row/column dependence.

Error/Issue if any
None
None
Exact SHM/motion subtopic not found in uploaded subtopic list.
None
None
Exact charged-pendulum equilibrium subtopic not found.
Exact volume-charge-density integration is not separately listed.
None
None
None
None
Typo: 'process' should be 'possess'. Radius 2 mm is unusually large for Millikan setup but data are usable.
None
None
None
Ambiguous wording: starts with charged particle of charge q but asks path of electron; sign convention is unclear.
None
Figure/text ambiguity: statement says charge C is -2q, while the figure appears to label C as +2q.
None
None
None
Exact electrostatic-mechanical equilibrium subtopic not found.
None
None
None
Typo/garbled symbol after 'is' in the question statement.
ppm basis is implicit; assumes dilute aqueous solution where 1 kg approximately equals 1 L.
None

None
None
None
None
None
None
Minor notation/capitalization issue: 'henrys law' should be 'Henry's law'.
None
None
None
None
None
Exact subtopic for classification by physical state of solute/solvent not found.
None
Missing punctuation before 'Hence'.
Chemical formula typo: 'HS2' should be 'H2S'.
None
Wording typo: 'upon dissolved' should be 'upon dissolving'.
Minor wording issue: missing connector after 'ideal solution'.
None
None
Grammar: 'increase' should be 'increases' in statement 1.
Question formatting is dense; determinant layout should be checked from original PDF image.
Exact trace/basic-matrix subtopic not found in uploaded list.
None
None
Matrix-power notation is visually dense but interpretable from the PDF.
Options B and C appear repeated/identical in the uploaded question paper.
None
None
None
None
None

None
None
Possible issue: condition a^2 + b^2 + c^2 = -2 requires complex values unless otherwise stated.
None
None
None
None
None
None
None
None
None
Exact Cayley-Hamilton/characteristic equation subtopic not found.
Determinant formatting is cramped; readable from the original PDF image.